From Virtual Machines to Docker Containers: The Evolution of Software Development

Containerisation and Kubernetes have eased software development, making it faster and better. Let's see where these are headed, looking at trends that are making life easier for developers.

Whether we're using a banking app or watching videos, we expect all software applications to run smoothly, no matter what device or platform we use. Technologies like virtualisation and containers, and orchestration tools such as Docker and Kubernetes have restructured how applications are built, tested, and deployed. They make it easier for developers to create and run software consistently across different environments.

In the past, software development faced a lot of challenges—compatibility issues, scaling problems, and time-consuming deployment. These old methods often led to the well-known phrase, "it works on my machine," when software didn't behave the same way in different environments. As applications grew more complex with many dependencies and varying setups, finding more reliable solutions became essential.

Virtualisation, introduced in the 1960s, revolutionised computing by enabling multiple operating systems to run on a single machine. In the 2000s, containerisation was a significant development, with tools like Linux containers (LXC). The concept gained widespread attention with the launch of Docker in 2013, which made containerisation easier and more accessible for developers. Docker made it easy for developers to bundle applications and their requirements into portable containers. This allowed isolated environments to be easily deployed anywhere, cutting down the overhead that came with traditional virtual machines. Building on Docker's success, Kubernetes arrived in 2014, created by Google to meet the need for managing containers at scale. With Kubernetes, developers gained powerful tools for automating the deployment, scaling, and management